GLOBAL RECIPROCAL COLLEGES | BS INFORMATION TECHNOLOGY

A Scheduling and Duty Management System

with AI Application in Absence Handling

CAPSTONE DEFENSE • KAIZEN SECURITY AGENCY CORP.

Nel Laurice T. Acop • Ranzie Kirth O. Cahulugan
Ashley Eunice V. Cruz • Christian Mark G. Regacho

Adviser: Mrs. Ann Camile M. Maupay

SCHED

Why the study was needed

Kaizen Security Agency Corp. • Security deployment across 25 LRT stations

25
LRT stations

15 days
per scheduling period

3 roles
Admin • Head Guard • Security

- Excel rosters required manual balancing of posts, shifts, and rest days.
- Unexpected absences triggered repeated searches and phone calls.
- Delayed replacements could extend the shifts of on-duty guards.

Study focus: connect roster preparation, absence handling, and duty information.

Source: manuscript, Ch. I; Ch. V, pp. 127–129

What the study aimed to achieve

General objective: develop a scheduling and duty management system with AI-assisted absence handling.

01 **Automate schedules**
Follow station needs, rest days, and rotation rules.

02 **Recommend replacements**
Screen candidates and add AI-assisted ranking.

03 **Streamline absence handling**
Support quicker assignments and reduce avoidable shift extensions.

04 **Evaluate the system**
Assess accessibility, timeliness, and usability.

Source: manuscript, Ch. I, pp. 5–6

What the system covers

Deployment: Windows and web for administrators; Android for guards and head guards.

Included

- 15-day schedules, rotation, and station setup
- Attendance reports, requests, and replacements
- In-app alerts, archives, and PDF reports

Outside the study scope

- Payroll and overtime-pay computation
- Biometric, GPS, or automatic time recording
- SMS, email, and iOS support

The AI recommends; the administrator confirms the final replacement.

Source: manuscript, Ch. I, pp. 6–11

Who benefits from the study

Expected value for agency operations and future development.

Agency & admins

- Centralized records and roster control
- Less manual work when staffing changes

Security personnel

- Duty details available on mobile
- Clear requests, updates, and cover assignments

Researchers

- Practical system-development experience
- Reference for AI-assisted workforce tools

The intended benefit is clearer coordination and more timely staffing decisions.

Source: manuscript, Ch. I, pp. 12–13

What the literature supports

Selected themes from the paper’s local and foreign literature review.

Local literature

- Workload-based staffing supports balanced assignments.
- AI adoption needs user readiness and responsible data handling.

Aytona et al. (2022) • Andaya (2025)

Foreign literature

- Security scheduling affects coverage and employee workload.
- AI can support complex workforce-planning decisions.

Hughes (2024) • Kupunarapu (2022)

Design implication: combine scheduling rules, usable interfaces, and human oversight.

Source: manuscript, Ch. II, pp. 17–23; citations as presented in the manuscript

The gap addressed by this project

Selected studies establish useful approaches; the project combines them for LRT absence handling.

Bulawat & Peña (2025)	Automated timetables and conflict detection
Gencer et al. (2024)	Metro-security placement with personnel preferences
Zupančič & Panov (2024)	Historical data for predicting employee absence

- Identified gap: integrated response to unexpected duty absences.
- Contribution: eligible shortlist → AI explanation → approved replacement.

Source: manuscript, Ch. II, pp. 26–34

Rapid Application Development

Working prototypes, stakeholder feedback, and repeated testing guided development.

01

Requirements

Identify users, rules, and station needs.

02

User design

Prototype interfaces and map workflows.

03

Construction

Build and refine working features.

04

Test & deploy

Validate functions and deploy clients.

Feedback and refinement across the development cycle

Why RAD: operational rules and interfaces could be refined through repeated feedback.

Source: manuscript, Ch. III, pp. 35–38; development timeline, pp. 89–92

Input → Process → Output

The system turns operational records into schedules and actionable duty information.

INPUT

- Personnel and availability
- Station posts and shifts
- Requests and absences

PROCESS

- Validate and apply rules
- Generate duty schedules
- Rank eligible replacements

OUTPUT

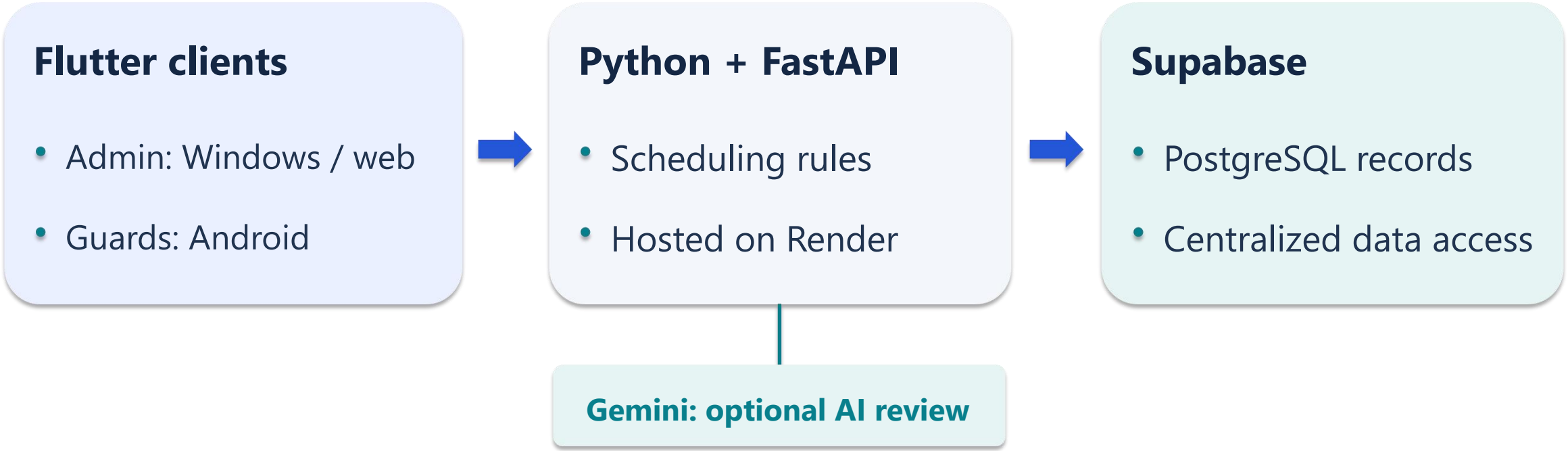
- Published duty rosters
- Approved replacements
- Reports and notifications

Administrative review remains part of publishing schedules and assigning replacements.

Source: manuscript, Ch. III, pp. 38–39; IPO redrawn for presentation

How the system is built

A shared backend connects desktop, web, and mobile clients.



Operating needs: a Windows PC or browser, Android devices, and internet access.

Three actors, one duty record

Context, data-flow, use-case, and entity diagrams define the same connected workflow.

Administrator

- Configure, generate, publish
- Approve requests and covers

Head Guard

- View the station roster
- Submit attendance per shift

Security personnel

- View personal duty schedules
- File requests and cover offers

Core data path Personnel + Stations → Schedules → Duty assignments → Attendance / Replacements

18 tables organize reference data, rotation, schedules, requests, and audit records.

Source: manuscript, Ch. III, pp. 46–69; simplified design summary

How the study evaluated the system

Descriptive-quantitative evaluation using an ISO/IEC 25010-based instrument.

5
IT expert respondents

211
end-user respondents

1–5
Likert rating scale

- Six criteria: functionality, efficiency, compatibility, reliability, usability, maintainability.
- Ratings summarized by criterion and by respondent group.
- 4.21–5.00 interpreted as “Highly Acceptable.”

Source: manuscript, Ch. III, pp. 93–98; Ch. IV, pp. 114–126

Human control and responsible data use

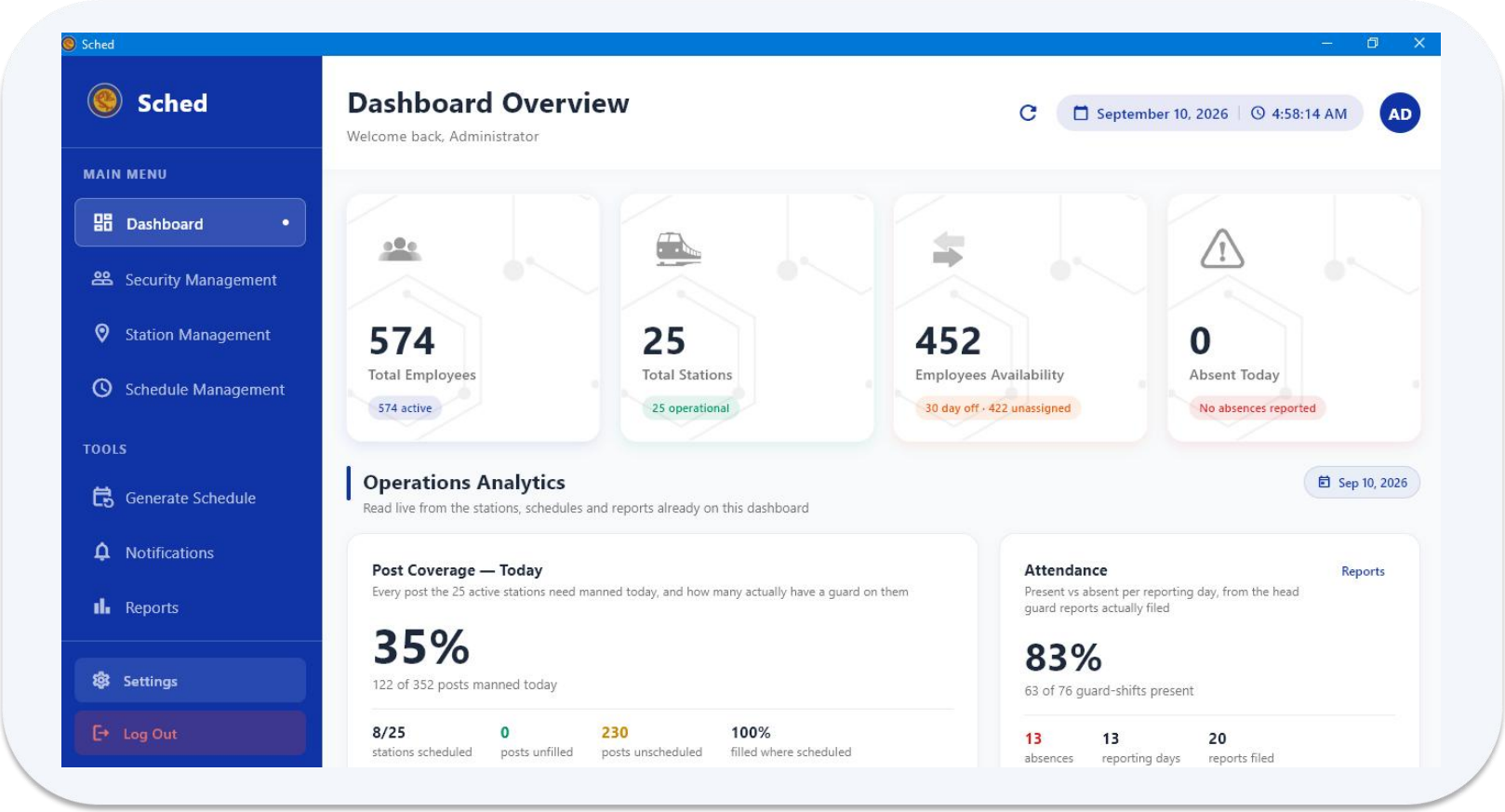
Safeguards described in the research method and reflected in the system design.

- Agency permission, informed consent, and voluntary participation.
- Evaluation responses reported in aggregate.
- Role-based access and hashed credentials.
- AI receives candidate identifiers and job attributes; direct contact details are excluded.
- The administrator decides; recommendations and assignments are recorded.

Source: manuscript, Ch. III, pp. 98–101; AI service and review modules

Sched: the administrator workspace

The implemented application connects roster preparation with day-to-day duty management.



- Monitor personnel and station coverage.
- Manage schedules, requests, and reports.
- Review staffing gaps before publication.

Source: manuscript, Appendix D, p. 155; screenshot reproduced from the paper

Automated schedules follow station requirements

The generator applies staffing rules, rest days, and station rotation.

1.7 ms
reported median: one station period

0.04 s
reported generation: all 25 stations

7,700
duty assignments in that benchmark

- Coverage follows configured shift and post requirements.
- Tests checked double bookings, rest days, and rotation behavior.
- Staffing shortfalls appear as notes for administrator review.

Benchmark scope: scheduling computation; full user response also includes data and network time.

Source: manuscript, Ch. IV, p. 104; Ch. V, pp. 127–128

How AI assists absence handling

Rules establish eligibility; Gemini adds a recommendation and explanation.

1

SCREEN

Check availability, conflicts, shift and post

2

RANK

Score familiarity, workload and offers

3

AI REVIEW

Compare candidates; explain the choice

4

CONFIRM

Admin approves; update roster and alerts

- AI output is checked against the eligible shortlist.
- If AI fails, the rule-based ranking remains available.

Source: manuscript, Ch. III, pp. 61–62; Ch. IV, p. 106; Ch. V, p. 128

A shorter replacement workflow

The system reduces manual coordination steps and provides a clear reporting time.

Previous process

- Review Excel and search for available guards
- Call candidates; repeat when they decline

Implemented process

- Receive absence and open a ranked shortlist
- Confirm a cover; update duty and notify guard

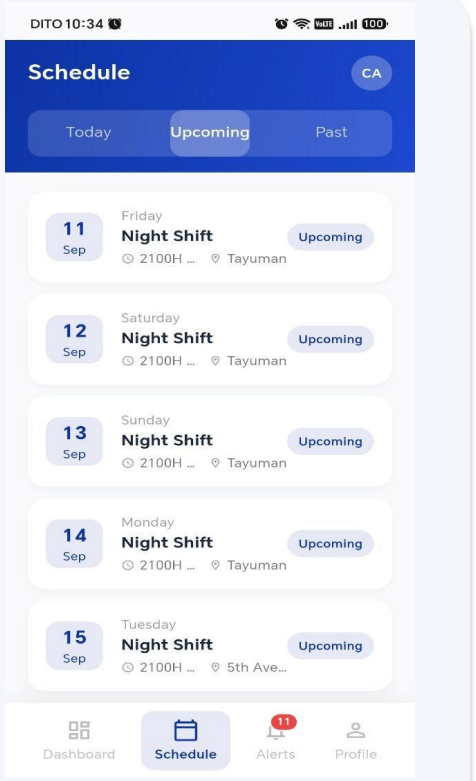
Example: 05:00–17:00 post + assignment at 09:00 → adjusted duty 10:00–19:00

Field reduction in replacement waiting time and overtime still requires before-and-after measurement.

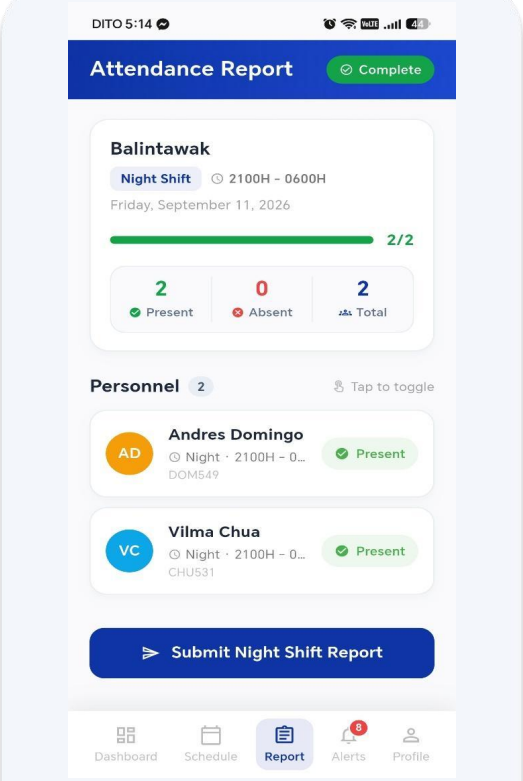
Source: manuscript, Ch. V, pp. 128–129, 133–135; preparation-window module

Duty information reaches personnel on mobile

Guard and head-guard views use the same shared operational records.



Personal schedule



Shift attendance report

- View current and past duties.
- File absence, leave, and day-off requests.
- Receive decisions and cover assignments.
- Head guards report attendance for their own shifts.

Source: manuscript, Appendix D, pp. 182, 192; Ch. IV, pp. 108–113

What the reported tests established

Evidence covers scheduling rules, API responses, and the deployed service.

284 / 284
automated backend checks passed

58 / 58
read endpoints answered

10 / 11
logged defects resolved

- Checks covered scheduling, requests, replacement rules, and AI fallback.
- Deployment testing returned HTTP 200 for the requested endpoints.
- One reported defect remained: narrow-window notification-tab overflow.

These are the paper’s reported test results; they do not mean every possible scenario is covered.

Source: manuscript, Ch. IV, pp. 102–113; Ch. V, p. 130

Both groups rated the system highly acceptable

Mean ratings on a five-point scale • IT experts (n = 5) • End users (n = 211)

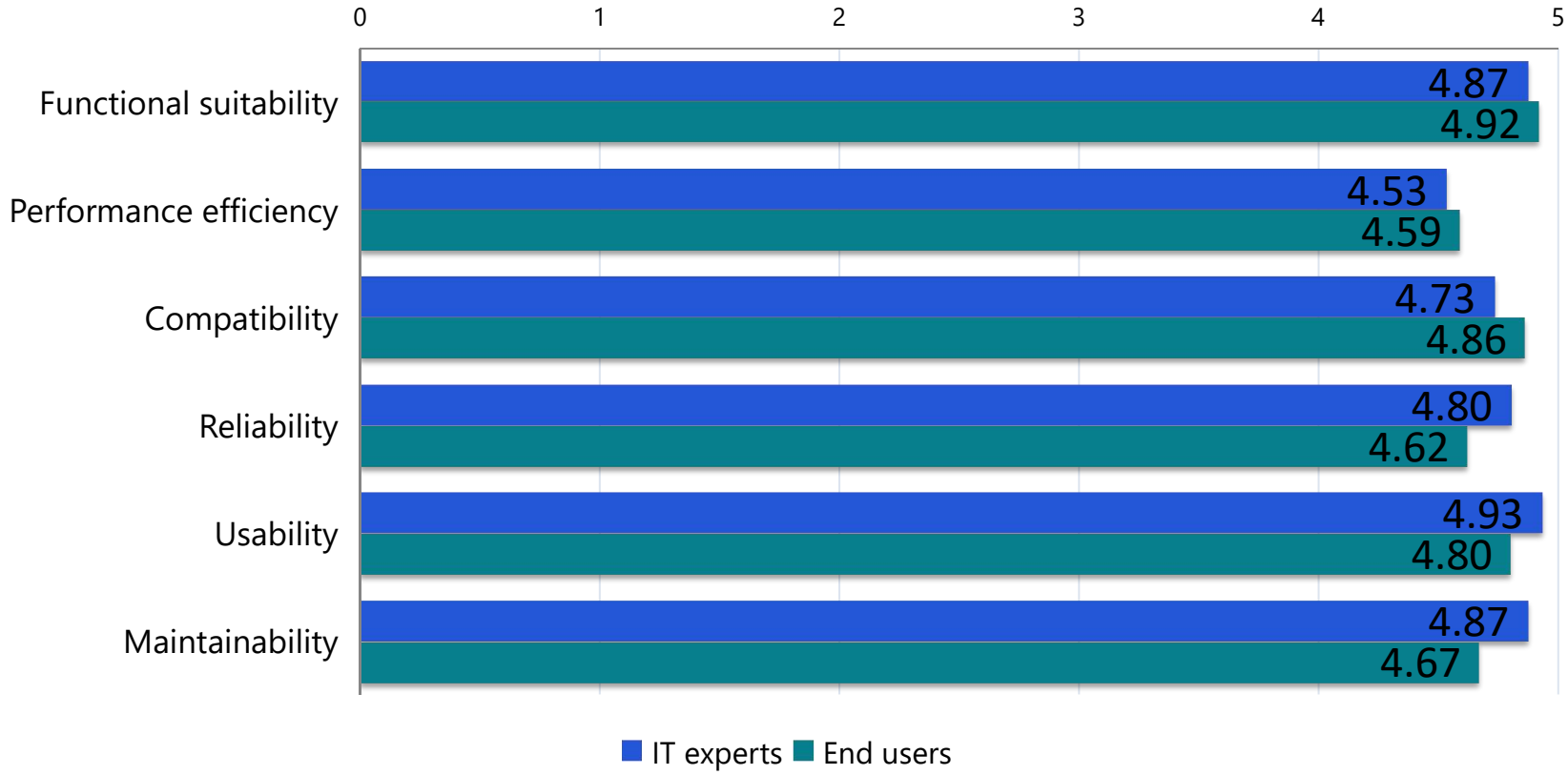
4.79 / 5

IT experts: overall rating

4.74 / 5

End users: overall rating

Efficiency rated lowest in both groups.



Source: manuscript, Table 26, p. 114; Table 33, p. 124; interpretation, pp. 125–126

Findings aligned with the objectives

The evidence combines implemented functions, reported tests, and user evaluation.

Scheduling	Station-based 15-day rosters; staffing gaps are visible.
Absence handling	Screened candidates, explained AI advice, and rule-based fallback.
Workflow	Less manual searching; field time savings remain to be measured.
User access	Shared duty access; high acceptability in both respondent groups.

Source: manuscript, Ch. V, pp. 127–130; Ch. IV evaluation tables

What the study concludes

The delivered system connects routine scheduling with day-to-day absence response.

- Automated roster generation reduces manual schedule construction.
- Rules and AI advice support administrator decisions on replacement candidates.
- Shared desktop, web, and mobile records improve access to duty information.
- Testing and respondent ratings support use in the agency’s studied setting.

Continuous coverage still depends on available personnel, timely reporting, and actual arrival.

Source: manuscript, Ch. V, pp. 131–133

What should happen next

Priorities drawn from the paper’s recommendations.

Agency operations

- Train users on requests and shift reporting.
- Review staffing notes and confirm AI suggestions.

System improvements

- Fix layout issues and hosting delays.
- Use staging tests; add offline and wider notification support.

Future research

- Measure replacement time, overtime, and satisfaction.
- Compare AI advice with rule-based ranking.

Priority: measure the operational benefits during sustained real-world use.

Source: manuscript, Ch. V, pp. 134–135

Thank you.

Questions and discussion

- Automated schedules
- AI-assisted replacements
- Administrator control